Appendix B

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1. Show that the unit disc in $\mathbb{C}$ is not isomorphic to X_h for any scheme X.

2. Let $z_1, z_2, \ldots$ be an infinite sequence of complex numbers with $|z_n| \to \infty$ as $n \to \infty$. Let $\mathfrak{J} \subset \mathcal{O}_{\mathbb{C}}$ be the sheaf of ideals of holomorphic functions vanishing at all of the z_n. Show that there is no coherent algebraic sheaf of ideals $\mathcal{I} \subset \mathcal{O}_X$, where $X = \mathbb{A}^1_{\mathbb{C}}$ is the affine line, such that $\mathfrak{J} = \mathcal{I}_h$ as an ideal in $\mathcal{O}_{\mathbb{C}}$. Show on the other hand that there is a coherent sheaf $\mathcal{F}$ on X such that $\mathcal{F}_h = \mathfrak{J}$ as coherent sheaves.

3. (Serre [12].) On $\mathbb{C}^2 - (0,0)$, we define an invertible analytic sheaf $\mathfrak{L}$ as follows: $\mathfrak{L} \cong \mathcal{O}$ when $z \neq 0$, $\mathfrak{L} \cong \mathcal{O}$ when $w \neq 0$, and when both $z, w \neq 0$, the two copies of $\mathcal{O}$ are glued by multiplication by $e^{-1/zw}$ in the local ring at the point (z, w). Show that there is no invertible algebraic sheaf $\mathcal{L}$ on $\mathbb{A}^2 - (0,0)$ with $\mathcal{L}_h \cong \mathfrak{L}$.

4. Show directly that if X is a scheme which is reduced and proper over $\mathbb{C}$, then $\mathrm{H}^0(X, \mathcal{O}_X) \cong \mathrm{H}^0(X_h, \mathcal{O}_{X_h})$. Conversely, show that if X is not proper over $\mathbb{C}$, then there is a coherent sheaf $\mathcal{F}$ on X with $\mathrm{H}^0(X, \mathcal{F}) \neq \mathrm{H}^0(X_h, \mathcal{F}_h)$.

5. If X, X' are nonsingular affine algebraic curves, with $X_h \cong X'_h$, show that $X \cong X'$.

6. Show that if X and Y are projective schemes over $\mathbb{C}$, and $\mathfrak{f} : X_h \to Y_h$ is a morphism of analytic spaces, then there exists a (unique) morphism $f : X \to Y$ with $f_h = \mathfrak{f}$.